

SQL Queries Documentation - Food Wastage Management System

Complete Query Reference Guide

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Overview

This document provides detailed explanations of all 15 SQL queries used in the Food Wastage Management System. Each query serves a specific business purpose in analyzing food donations, tracking claims, and optimizing food distribution.

Database Schema

- Providers (Provider_ID, Name, Type, Address, City, Contact)
- Receivers (Receiver_ID, Name, Type, City, Contact)
- Food_Listings (Food_ID, Food_Name, Quantity, Expiry_Date, Provider_ID, Provider_Type, Location, Food_Type, Meal_Type)
- Claims (Claim_ID, Food_ID, Receiver_ID, Status, Timestamp)

Category 1: Providers & Receivers

Q1: Providers and Receivers Count by City

Purpose:

- Understand the distribution of food providers and receivers across different cities
- Identify cities with supply-demand imbalances
- Plan resource allocation and outreach programs

Business Value:

- Strategic planning for expansion
- Identify underserved cities
- Balance supply and demand

Use Cases:

- Dashboard KPI display
- Geographic expansion planning
- Marketing campaign targeting

Q2: Provider Type Contributions

Purpose:

- Identify which type of provider (Restaurant, Grocery Store, Supermarket) contributes the most food
- Recognize top contributing segments
- Focus partnership efforts

Business Value:

- Optimize partnership strategies
- Recognize and reward top contributors
- Target recruitment efforts

Use Cases:

- Provider performance reports
- Partnership development
- Incentive program design

Q3: Provider Contact Information

Purpose:

- Provide easy access to provider contact details
- Enable direct communication between receivers and providers
- Facilitate coordination for food pickup

Business Value:

- Streamline communication
- Reduce coordination time
- Improve claim fulfillment rates

Use Cases:

- Contact directory in app
- Provider lookup by city
- Emergency food sourcing

Q4: Top Receivers by Food Claimed

Purpose:

- Identify the most active receivers
- Understand food distribution patterns
- Recognize organizations making the most impact

Business Value:

- Acknowledge impactful partners
- Ensure equitable distribution
- Detect potential misuse

Use Cases:

- Impact reports
- Partner recognition programs
- Distribution fairness analysis

Category 2: Food Listings & Availability

Q5: Total Food Available

Purpose:

- Track real-time food availability across the system
- Monitor unclaimed food inventory
- Prevent food wastage

Business Value:

- Real-time inventory visibility
- Wastage prevention
- Resource optimization

Use Cases:

- Dashboard main metrics
- Daily inventory reports
- Alert system for unclaimed food

Q6: Cities with Most Food Listings

Purpose:

- Identify cities with highest food availability
- Compare food supply across locations
- Guide receiver targeting

Business Value:

- Supply hotspot identification
- Geographic distribution insights
- Targeted receiver recruitment

Use Cases:

- Geographic performance dashboards
- City comparison reports
- Resource allocation planning

Q7: Most Common Food Types

Purpose:

- Understand what types of food are most commonly available
- Match receiver needs with available food types
- Identify dietary preference trends

Business Value:

- Menu planning for receivers
- Food type demand forecasting
- Dietary requirement matching

Use Cases:

- Food type filtering in app
- Dietary preference reports
- Provider guidance on demand

Category 3: Claims & Distribution

Q8: Claims per Food Item

Purpose:

- Track popularity and demand for specific food items
- Identify high-demand vs. low-demand food
- Optimize future food listings

Business Value:

- Demand pattern analysis
- Inventory optimization
- Wastage reduction through insights

Use Cases:

- Popular food item reports
- Demand forecasting
- Provider recommendations

Q9: Providers with Highest Successful Claims

Purpose:

- Recognize most successful providers
- Identify best practices
- Build provider leaderboard

Business Value:

- Provider recognition programs
- Best practice sharing
- Partnership strengthening

Use Cases:

- Provider leaderboards
- Performance awards
- Case study identification

Q10: Claim Status Distribution

Purpose:

- Monitor claim success rates
- Identify bottlenecks in distribution
- Track system efficiency

Business Value:

- Process improvement insights
- Operational efficiency metrics
- Quality assurance

Use Cases:

- System health monitoring
- Process optimization
- Success rate KPIs

Category 4: Analysis & Insights

Q11: Average Quantity per Receiver

Purpose:

- Understand typical claim sizes by receiver
- Identify outliers (very large or small claims)
- Ensure fair distribution

Business Value:

- Fair distribution monitoring
- Receiver capacity planning
- Anomaly detection

Use Cases:

- Fair distribution reports
- Receiver capacity analysis
- Fraud detection

Q12: Most Claimed Meal Types

Purpose:

- Understand which meals are in highest demand
- Optimize provider meal planning
- Match supply with demand patterns

Business Value:

- Demand-driven supply planning
- Provider guidance
- Time-based distribution optimization

Use Cases:

- Meal planning guidance
- Time-based alerts
- Provider recommendations

Q13: Total Donations by Provider

Purpose:

- Track individual provider contribution levels
- Build provider performance profiles
- Create donation leaderboards

Business Value:

- Provider recognition
- Donation tracking
- Impact measurement

Use Cases:

- Annual impact reports
- Provider certificates
- Donation history tracking

Category 5: Additional Queries

Q14: Expiring Food Alert

Purpose:

- Identify food items expiring within 3 days
- Enable urgent distribution
- Minimize food wastage

Business Value:

- Wastage prevention
- Urgent action triggers
- Real-time alerts

Use Cases:

- Urgent alert system
- Priority distribution queue
- Wastage prevention dashboard

Q15: City Efficiency Metrics

Purpose:

- Comprehensive city-level performance analysis
- Compare operational efficiency across cities
- Identify best and worst performing locations

Business Value:

- Strategic city-level planning
- Resource allocation optimization
- Performance benchmarking

Use Cases:

- City comparison dashboards
- Performance scorecards
- Strategic planning reports

Query Usage Matrix

For Providers:

- Q2: See contribution rankings
- Q9: Track successful distributions
- Q13: View donation history
- Q14: Monitor expiring inventory

For Receivers:

- Q3: Find provider contacts
- Q5: Check food availability
- Q7: Filter by food type
- Q12: Find preferred meal types

For Administrators:

- Q1: Overview of system coverage
- Q6: Geographic performance
- Q10: System health metrics
- Q15: Comprehensive city analysis

For Data Analysis:

- Q8: Demand forecasting
- Q11: Distribution fairness
- Q12: Temporal patterns
- Q13: Impact measurement